

$\mathcal{N}=4$ Super Yang-Mills Theory

DRP 2026

Yusuf Berk AKÇAY

28 June 2026

Outline

- 1 Causality and Locality
- 2 Real Scalar Field & Perturbation
- 3 Renormalization
- 4 Yang-Mills Theory
- 5 Supersymmetry (SUSY)
- 6 $\mathcal{N} = 4$ Super Yang-Mills Theory

Why Causality Leads Us to QFT

- In standard QM, assume a localized state $|\vec{x}\rangle$ and two spacetime points $(0, 0)$ and $(t, \vec{x})$.
- The transition amplitude between these points is:

$$A = \langle \vec{x} | e^{-itH/\hbar} | \vec{x} = 0 \rangle = \int \frac{d^3p}{(2\pi)^3} e^{-ix^\mu p_\mu / \hbar} \neq 0$$

- **The Issue:** Even if the points are strictly spacelike separated (outside the lightcone), the amplitude does not vanish!
- **Conclusion:** Localized states and a definitive position operator are incompatible with relativity.
- **The Solution:** We must view time and space not as observables, but as *labels* for operators.

$$QM \rightarrow QFT \quad \implies \quad \mathcal{O}(t) \rightarrow \mathcal{O}(x^\mu)$$

Real Scalar Field & Quantization

- The simplest Poincare invariant construction is the real scalar field:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \quad \Longrightarrow \quad (\square + m^2) \phi = 0$$

- **Quantization (CCR):** $[\phi(t, \vec{x}), \pi(t, \vec{y})] = i\delta^3(\vec{x} - \vec{y})$
- **Fourier Expansion:**

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k e^{-ikx} + a_k^\dagger e^{ikx} \right)$$

- **Fock Space:** Defined by a unique, Lorentz invariant vacuum $a_k |0\rangle = 0$. Multi-particle states are built using creation operators $a_k^\dagger$.

Perturbation Theory

- For interacting theories $H = H_0 + gH_{int}$ ($g \ll 1$), exact solutions are generally impossible. We use the **Perturbation Theory**.
- Time evolution is governed by the **Dyson Formula**:

$$U(t, t_0) = T \exp \left(-i \int_{t_0}^t H_I(t') dt' \right)$$

- Time-ordered correlation functions of interacting fields are given by:

$$\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle T \phi(x_1) \dots \phi(x_n) \exp(i \int d^4x \mathcal{L}_{int}) \rangle_0}{\langle T \exp(i \int d^4x \mathcal{L}_{int}) \rangle_0}$$

- If we expand the above formula, we can represent each term by graphs, called the **Feynman Diagrams**.

Renormalization (1): The Motivation & The 3-Step Program

The Motivation: The Infinity Problem

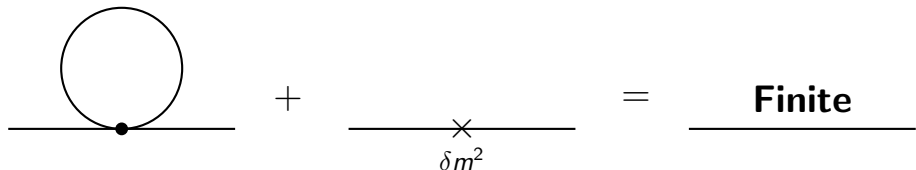
In QFT, virtual particles in loop diagrams can possess arbitrary momenta k up to infinity. Integrating over these internal momenta leads to divergent (infinite) results for physical observables like mass or charge.

The 3-Step Solution:

- 1 **Regularization:** Modify the theory temporarily to make integrals finite. For example, introduce a highly energetic "Cutoff" scale Λ (integrating only up to Λ).
- 2 **Renormalization:** Absorb the Λ -dependent infinities into the unobservable, "bare" parameters of the Lagrangian by adding *counterterms*.
- 3 **Removing the Cutoff:** Take the physical limit ($\Lambda \rightarrow \infty$). The remaining observable physical quantities are perfectly finite and independent of Λ !

Renormalization (2): Visualizing Mass Renormalization in ϕ^4

In $V = \lambda\phi^4/4!$ theory, the 1-loop self-energy modifies the particle's mass. With a cutoff Λ , the loop integral diverges as $\sim \Lambda^2$.



The diagram shows the renormalization of the mass in ϕ^4 theory. On the left, a horizontal line with a black dot at its center has a circle (loop) attached to the dot. This is followed by a plus sign. To the right of the plus sign is another horizontal line with a cross (X) in the middle. Below the cross is the label δm^2 . This is followed by an equals sign. To the right of the equals sign is a single horizontal line. To the right of this line is the word "Finite".

- We split the original Lagrangian parameters into physical parameters + counterterms:
 $\mathcal{L} \supset -\frac{1}{2}m_{bare}^2\phi^2 = -\frac{1}{2}m_{phys}^2\phi^2 - \frac{1}{2}\delta m^2\phi^2$
- We intentionally choose the counterterm $\delta m^2 \sim \Lambda^2$ to perfectly cancel the loop divergence.
- **Result:** $m_{phys}^2 = m_{bare}^2 + \Sigma_{loop} = \text{Finite}$ (measurable mass).

Yang-Mills Theory: Localizing Symmetries

- Let's promote a global symmetry $\psi(x) \mapsto h\psi(x)$ to a **local symmetry**:

$$\psi(x) \mapsto h(x)\psi(x)$$

- The Issue:** Normal derivatives break covariance because they compare fields at different spacetime points:

$$\partial_\mu \psi(x) \mapsto h(x)\partial_\mu \psi(x) + \psi(x)\partial_\mu h(x)$$

- Solution (Wilson Line):** Introduce $C(y, x)$ to properly compare fields. This naturally defines the **Covariant Derivative**:

$$D_\mu \psi(x) = \partial_\mu \psi(x) + igA_\mu(x)\psi(x) \quad \implies \quad D_\mu \psi(x) \mapsto h(x)D_\mu \psi(x)$$

- Thus, the Gauge Field $A_\mu(x)$ is inevitably born!

Yang-Mills Theory: Field Strength & Action

- The gauge field $A_\mu(x)$ takes values in the Lie algebra. To provide dynamics, we define the **Field Strength Tensor**:

$$F_{\mu\nu} = \frac{1}{ig}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]$$

- The field strength transforms covariantly:

$$F_{\mu\nu}(x) \mapsto h(x)F_{\mu\nu}(x)h^{-1}(x)$$

- We can now construct a gauge-invariant Lagrangian term by taking the trace:

$$\mathcal{L}_{YM} = -\frac{1}{2}\text{tr}(F_{\mu\nu}F^{\mu\nu})$$

The Coleman-Mandula Theorem & Its Loopholes

Coleman-Mandula Theorem (1967)

In any interacting QFT with a mass gap, the most general Lie algebra of symmetries of the S-matrix is the direct sum of the Poincare algebra and an internal symmetry algebra:

$$\text{Total Symmetry} = \text{Poincare} \oplus \text{Internal} \implies [P_\mu, T_a] = 0, \quad [M_{\mu\nu}, T_a] = 0$$

How to evade this “no-go” theorem?

- **Loophole 1 (Supersymmetry):** The theorem assumes a standard Lie algebra. SUSY uses *graded* Lie algebras (anti-commutators) to mix spacetime and internal properties (fermions $\leftrightarrow$ bosons).
- **Loophole 2 (Conformal Symmetry):** The theorem assumes a *mass gap*. In some massless theories, Poincare extends to the **Conformal Group** $SO(4, 2)$, naturally mixing scale and momentum ($[D, P_\mu] = -iP_\mu$).

$\implies \mathcal{N} = 4$ SYM uses **both** loopholes to form a Superconformal Algebra!

SUSY Algebra & Representations

- The Coleman-Mandula theorem restricts symmetries. SUSY bypasses it using *graded* Lie algebras.
- $\mathcal{N} = 1$ SUSY Algebra:

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma_{\alpha\dot{\beta}}^\mu P_\mu, \quad \{Q_\alpha, Q_\beta\} = 0, \quad \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0$$

- **Representations (Supermultiplets):**

- Since $[P^2, Q_\alpha] = 0$, all particles in a supermultiplet have the **same mass**.
- The fermion number operator trace $Tr(-1)^F = 0$ implies an **equal number of bosons and fermions**.
- **Massless Representation ($P^2 = 0$):** From a vacuum $\lambda_0 = 0$, we get a supermultiplet, called chiral supermultiplet, with helicities $\{0, 1/2\}$. Including its CPT conjugate yields $\{|0\rangle, |0'\rangle, |-1/2\rangle, |1/2\rangle\}$.

Superspace Formalism

- Poincare-invariant Lagrangians are most conveniently defined in the Minkowski space, which has the Poincare symmetry. Similarly, SUSY-invariant Lagrangians are most conveniently defined in the **Superspace**, which has the SUSY symmetry.
- We exponentiate the Super-Poincare algebra to define **Superspace**:

$$\text{Superspace} = \frac{\text{Super-Poincare Group}}{\text{Lorentz Group}} = \{(x^\mu, \theta^\alpha, \bar{\theta}^{\dot{\alpha}})\}$$

where $\theta, \bar{\theta}$ are anti-commuting Grassmannian numbers.

- A **Superfield** $Y(x, \theta, \bar{\theta})$ encodes a supermultiplet. The Taylor expansion terminates naturally because $\theta\theta\theta = 0$:

$$Y = \phi(x) + \theta\psi(x) + \bar{\theta}\bar{\chi}(x) + \cdots + \theta\theta\bar{\theta}\bar{\theta}D(x)$$

- **Irreducible Superfields:**

- *Chiral Superfield* (Φ): Defined by $\bar{D}_{\dot{\alpha}}\Phi = 0$.
- *Vector Superfield* (V): Defined by the reality condition $V^\dagger = V$.

SUSY Invariant Actions

- The highest θ -components of superfields (F-term for Chiral, D-term for Vector) transform as **total spacetime derivatives**.
- Therefore, integrals over superspace automatically yield **SUSY invariant actions**!

$$S = \int d^4x \left[\int d^2\theta d^2\bar{\theta} \Phi^\dagger \Phi + \int d^2\theta W(\Phi) + \int d^2\bar{\theta} W^\dagger(\Phi^\dagger) \right]$$

- With $W(\Phi) = m\Phi^2/2 + \lambda\Phi^3/3$, we obtain the **Wess-Zumino Action**:

$$S = \int d^4x \left[\partial_\mu \phi^\dagger \partial^\mu \phi - i\bar{\psi} \bar{\sigma}^\mu \partial_\mu \psi - \left| \frac{\partial W}{\partial \phi} \right|^2 - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^2} \psi \psi - h.c. \right]$$

Field Content of $\mathcal{N} = 4$ SYM

- $\mathcal{N} = 4$ Super Yang-Mills is the maximally supersymmetric theory in 4 dimensions (without gravity).
- All fields take values in the **adjoint representation** of the gauge group $SU(N)$ (i.e., $\mathfrak{su}(N)$ algebra).
- The field content consists of:
 - ① **1 Gauge Field (Gluon):** $A_\mu(x)$
 - ② **6 Real Scalar Fields:** $\phi^M(x)$ where $M = 1, \dots, 6$. (Often repacked as antisymmetric $SU(4)$ tensors $\phi^{AB}, \bar{\phi}_{AB}$).
 - ③ **4 Complex Weyl Fermions:** $\psi_\alpha^A(x)$ and their conjugates $\tilde{\psi}_{\dot{A}}^{\dot{\alpha}}(x)$ where $A \in \{1, 2, 3, 4\}$.
- *Question:* Constructing a fully $\mathcal{N} = 4$ supersymmetric Lagrangian by hand in 4D is incredibly hard. Is there a more elegant way?

The 10-Dimensional Origin: Dimensional Reduction

- **Solution:** We start with the simplest supersymmetric gauge theory, $\mathcal{N} = 1$ **SYM in 10 dimensions** (Lorentz Space $\mathbb{R}^{1,9}$).
- The 10D Lagrangian is remarkably compact and elegant:

$$\mathcal{L}_{10} = \text{tr} \left(-\frac{1}{2} F_{MN} F^{MN} + i \bar{\Psi} \Gamma_M D^M \Psi \right) \quad (M, N = 0, \dots, 9)$$

- **Dimensional Reduction:** Assume that the fields do not depend on the extra 6 spatial dimensions. ($\mathbb{R}^6 \rightarrow T^6 \rightarrow *$)
Thus, derivatives $\partial_a = 0$ for $a \in \{4, \dots, 9\}$.
- The 10D Gauge field A_M splits beautifully:
 - A_μ for $\mu = 0, 1, 2, 3 \implies$ The **4D Gauge Field** (Gluon).
 - A_{a+3} for $a = 1, \dots, 6 \implies$ The **6 Scalar Fields** ϕ^a .

Breaking Down the Bosonic Part

- Let's expand the 10D field strength tensor squared: $-\frac{1}{2}F_{MN}F^{MN}$.
- It naturally decomposes into three distinct 4D parts:

$$-\frac{1}{2}F_{MN}F^{MN} = \underbrace{-\frac{1}{2}F_{\mu\nu}F^{\mu\nu}}_{\text{4D Spacetime}} \underbrace{-F_{\mu a}F^{\mu a}}_{\text{Mixed}} \underbrace{-\frac{1}{2}F_{ab}F^{ab}}_{\text{Extra Dimensions}}$$

- **1. Gauge Kinetic Term:** $-\frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ remains the standard 4D Yang-Mills term.
- **2. Scalar Kinetic Term:** Since $\partial_a = 0$, $F_{\mu a} = \partial_\mu A_a - \partial_a A_\mu - ig[A_\mu, A_a] = D_\mu \phi_a$.

$$-F_{\mu a}F^{\mu a} = \frac{1}{2}(D_\mu \phi^{AB})(D^\mu \bar{\phi}_{AB})$$

- **3. Scalar Potential:** Since $\partial_a = \partial_b = 0$, $F_{ab} = -ig[A_a, A_b] = -ig[\phi_a, \phi_b]$.

$$-\frac{1}{2}F_{ab}F^{ab} = \frac{1}{8}g^2[\phi^{AB}, \phi^{CD}][\bar{\phi}_{AB}, \bar{\phi}_{CD}]$$

Fermions and the Final 4D Lagrangian

- The 32-component 10D Majorana-Weyl spinor Ψ decomposes exactly into the 4 pairs of 4D Weyl spinors $\psi_\alpha^A, \tilde{\psi}_A^{\dot{\alpha}}$.
- The 10D Dirac term $i\bar{\Psi}\Gamma_M D^M \Psi$ splits into:
 - **Fermion Kinetic terms** from the 4D spacetime components $(\Gamma_\mu D^\mu)$.
 - **Yukawa Interaction terms** from the extra dimensions, where $D_a \Psi = -ig[\phi_a, \Psi]$.
- **The Result:** Summing all parts, the full $\mathcal{N} = 4$ SYM Lagrangian emerges naturally:

$$\begin{aligned} \mathcal{L}_4 = \text{tr} \bigg(& -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} (D_\mu \phi^{AB})(D^\mu \bar{\phi}_{AB}) + \frac{1}{8} g^2 [\phi^{AB}, \phi^{CD}][\bar{\phi}_{AB}, \bar{\phi}_{CD}] \\ & + 2i\tilde{\psi}_{\dot{\alpha}A} \sigma_\mu^{\dot{\alpha}\beta} D^\mu \psi_\beta^A - \sqrt{2} g \psi^{\alpha A} [\bar{\phi}_{AB}, \psi_\alpha^B] + \sqrt{2} g \tilde{\psi}_{\dot{\alpha}A} [\phi^{AB}, \tilde{\psi}_B^{\dot{\alpha}}] \bigg) \end{aligned}$$

Why $\mathcal{N} = 4$ SYM? (The Harmonic Oscillator of the 21st Century)

- **UV Finiteness ($\beta = 0$):** Unlike QED or QCD, its β -function is exactly zero. The coupling constant doesn't run with energy. It is a completely finite theory at the quantum level!
- **AdS/CFT Correspondence:** It is the gauge theory side of Maldacena's duality. $\mathcal{N} = 4$ SYM in 4D is exactly equivalent to Type IIB String Theory in an $AdS_5 \times S^5$ background.
- **Integrability:** The theory is highly constrained. Operators map to spin chains in condensed matter physics, allowing us to compute exact dimensions at any coupling strength.
- **Scattering Amplitudes & Wilson Loops:** Its extreme symmetry reveals hidden mathematical structures (Twistors, Amplituhedron). Scattering amplitudes are surprisingly dual to **Null Polygonal Wilson Loops!**

Thank You!

Questions?